

Final Exam Revision Solution

1. Refer to the lecture slides

2. - Low Latency

Since the path is established before data transmission, there's no delay in data transmission.

- Guaranteed Bandwidth

A dedicated path is established before any data is transmitted, and this path remains reserved for the duration of the connection.

3.

Propagation delay

■ $d_{prop} = d/s$ (d : length of physical link, s : propagation speed)

$$\frac{2500 \times 10^3}{2.5 \times 10^8} = 10 \text{ msec}$$

Transmission delay

■ $d_{trans} = L/R$ (L : packet length (bits), R : link *bandwidth* (bps))

$$\frac{2000 \times 8}{100 \times 10^6} = 0.16 \text{ msec}$$

4. An ARP query is sent in a broadcast frame because the querying host does not know which adapter address corresponds to the IP address in question. For the response, the sending node knows the adapter address to which the response should be sent, so there is no need to send a broadcast frame (which would have to be processed by all the other nodes on the LAN).

5.

- i. from PC1 to R1:
Source MAC address: 54-89-98-4D-14-DD
Destination MAC address: 00e0-fc39-7ffe
Source IP: 10.1.1.2
Destination IP: 10.3.3.2
- ii. from R1 to R2:
Source MAC address: 00e0-fc39-7ffd

Destination MAC address: 00e0-fc90-11d3

Source IP: 10.1.1.2

Destination IP: 10.3.3.2

iii. from R2 to PC3:

Source MAC address: 00e0-fc90-11d4

Destination MAC address: 54-89-98-7E-29-AF

Source IP: 10.1.1.2

Destination IP: 10.3.3.2

6.

Action	Switch Table State	Link(s) packet is forwarded to	Explanation
B sends a frame to E	Switch learns interface corresponding to MAC address of B	A, C, D, E, and F	Since switch table is empty, so switch does not know the interface corresponding to MAC address of E
E replies with a frame to B	Switch learns interface corresponding to MAC address of E	B	Since switch already knows interface corresponding to MAC address of B
A sends a frame to B	Switch learns the interface corresponding to MAC address of A	B	Since switch already knows the interface corresponding to MAC address of B
B replies with a frame to A	Switch table state remains the same as before	A	Since switch already knows the interface corresponding to MAC address of A

7.

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address	Subnet Mask
1	130.56.0.0/26	130.56.0.1	130.56.0.62	130.56.0.63	255.255.255.192
2	130.56.0.64/26	130.56.0.65	130.56.0.126	130.56.0.127	255.255.255.192
3	130.56.0.128/26	130.56.0.129	130.56.0.190	130.56.0.191	255.255.255.192
4	130.56.0.192/26	130.56.0.193	130.56.0.254	130.56.0.255	255.255.255.192

8.

Initial:

X's table		
Destination	Cost	Next Hop
X	0	-
Y	2	Y
Z	4	Z

Y's table		
Destination	Cost	Next Hop
X	2	X
Y	0	-
Z	1	Z

Z's table		
Destination	Cost	Next Hop
X	4	X
Y	1	Y
Z	0	-

After 1st Iteration:

X's table		
Destination	Cost	Next Hop
X	0	-
Y	2	Y
Z	3	Y

Y's table		
Destination	Cost	Next Hop
X	2	X
Y	0	-
Z	1	Z

Z's table		
Destination	Cost	Next Hop
X	3	Y
Y	1	Y
Z	0	-

9.

- i) 3974188231 (x + 1)
- ii) 3974188231
- iii) 1884869394
- iv) 3974188231
- v) 3974188562 (3974188231+331)

10.

N'	D(u), p(u)	D(v), p(v)	D(w), p(w)	D(x), p(x)	D(y), p(y)	D(z), p(z)
t	2, t	4, t	∞	∞	7, t	∞
tu		4, t	5, u	∞	7, t	∞
tuv			5, u	7, v	7, t	∞
tuvw				7, v	7, t	∞
tuvwx					7, t	15, x
tuvwxy						15, x
tuvwxyz						

11.

- i) In the second segment from Host A to B, the sequence number is 207, source port number is 302 and destination port number is 80.

First segment sequence number (127) + bytes of data (80)

- ii) If the first segment arrives before the second, in the acknowledgement of the first arriving segment, the acknowledgement number is 207, the source port number is 80 and the destination port number is 302.

- iii) If the second segment arrives before the first segment, in the acknowledgement of the first arriving segment, the acknowledgement number is 127, indicating that it is still waiting for bytes 127 and onwards.

12.

- i) The document request was `http://gaia.cs.umass.edu/cs453/index.html`. The `Host:` field indicates the server's name and `/cs453/index.html` indicates the file name.
- ii) The browser is running HTTP version 1.1, as indicated in the first line of the HTTP request message.
- iii) The browser is requesting a persistent connection, as indicated by the `Connection: keep-alive`.
- iv) Mozilla/5.0. The browser type information is needed by the server to send different versions of the same object to different types of browsers.

13.

- i) The status code of 200 and the phrase OK indicate that the server was able to locate the document successfully. As indicated in the “Date” field of the HTTP response message, the reply was provided on Tuesday, 07 Mar 2008 12:39:45 Greenwich Mean Time.
- ii) As indicated in the “Last-Modified” field of the HTTP response message, the document `index.html` was last modified on Saturday 10 Dec 2005 18:27:46 GMT.
- iii) As indicated in the “Content-Length” field of the HTTP response message, there are 3874 bytes in the document being returned.
- iv) The first five bytes of the returned document are: `<!doc`.